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#1 Holy Angel Avenue, Sto. Rosario Angeles City, Pampanga Philippines 2009
SCHOOL OF COMPUTING

CASE STUDY FORM

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Case Title: Convert Number to Words	Section: iCT-107

NUMBER TO WORD CONVERTER

Narrative

Converting numbers to words may seem easy, but it is harder. The problem involves an inputted number that will turn into an English word that represents the value of the number. The hard part I've noticed is the more you keep adding place values, the more it gets complicated. Thankfully, it only stops at trillions based on the instructions. Moreover, the English word of the number must be correct and accurate. For example, turning the number '153,345' must be like this: "One Hundred Fifty-Three Thousand Three Hundred Forty-Five." Overall, the program challenges us to convert very large numbers into their word representation.

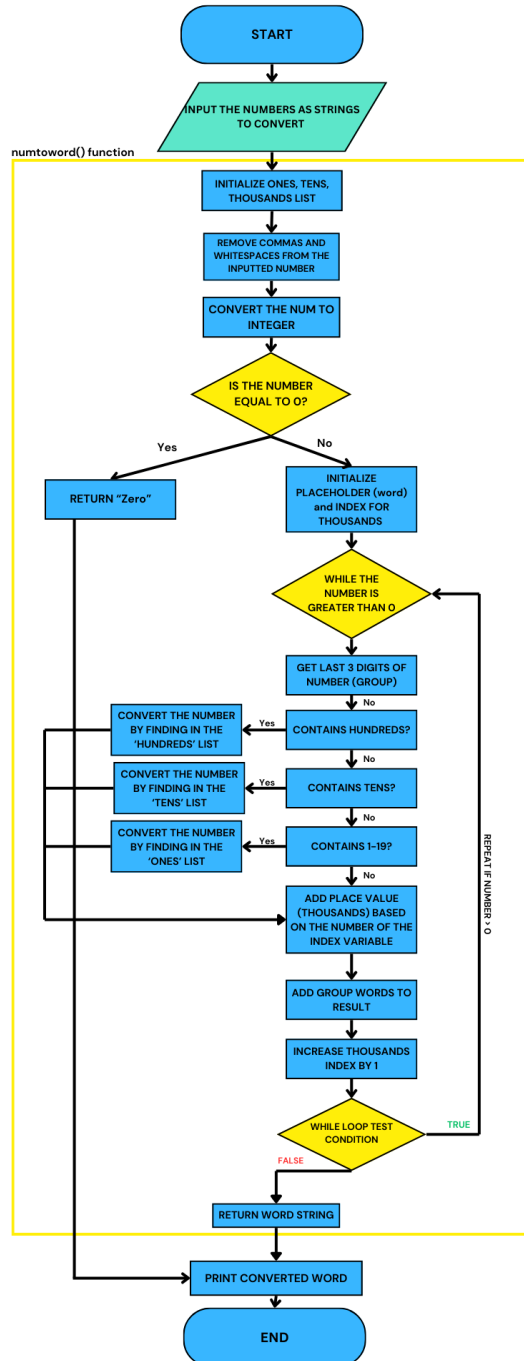
It took a while to understand the logic behind it, but the solution I have found is to make word arrays for the place values like ones, tens, hundreds, thousands, millions, billions, and trillions. However, the ones must contain word numbers ranging from 1 to 19 because each number has a unique name. The word arrays will be used as an access point whenever the number is whether in the ones, tens, hundreds, etc.

Furthermore, to convert them properly, my idea is to group the numbers by 3 digits, because going all out at once will take longer, and a long list of word arrays, especially since the program needs trillions so going all out at once instead of chunking is not efficient. After I manage to group the numbers, I will have decision paths whether the 3 digits are in ones, tens, or hundreds, and if the number is whether it is in those values, it will access the specific word array and append it to the word. Also, the solution keeps track of each group's placement with the index every time the while loop is iterated. Once that's done, it will print the converted word.

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Analysis



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Conclusion

Here's the snapshot of the full code:

```
1  def numtoword(num):
2      # Setting the Main Logic
3      ones = ['', 'One', 'Two', 'Three', 'Four', 'Five', 'Six', 'Seven', 'Eight', 'Nine', 'Ten',
4              'Eleven', 'Twelve', 'Thirteen', 'Fourteen', 'Fifteen', 'Sixteen', 'Seventeen',
5              'Eighteen', 'Nineteen']
6      tens = ['', 'Twenty', 'Thirty', 'Forty', 'Fifty', 'Sixty', 'Seventy', 'Eighty', 'Ninety']
7      hundreds = ['', 'One Hundred', 'Two Hundred', 'Three Hundred', 'Four Hundred', 'Five Hundred',
8                  'Six Hundred', 'Seven Hundred', 'Eight Hundred', 'Nine Hundred']
9      thousands = ['', 'Thousand', 'Million', 'Billion', 'Trillion']
10
11     # Conversion Starts
12     conversion = num.replace(",", "").replace(" ", "")
13     number = int(conversion)
14
15     if number == 0:
16         return "Zero"
17
18     word = ""
19     thousands_index = 0
20
21     while number > 0:
22         group = number % 1000
23         number //= 1000
24
25         if group > 0:
26             word_cluster = ""
27
28             if group >= 100: # 100 - 900
29                 word_cluster += hundreds[group // 100] + " "
30                 group %= 100
31
32             if group >= 20: # 20 - 90
33                 word_cluster += tens[group // 10] + " "
34                 group %= 10
35
36             if group > 0: # 1 - 19
37                 if group < 10: # It's a single digit number (1-9)
38                     if word_cluster.strip().endswith("Hundred"): # Check if it follows hundreds
39                         word_cluster += ones[group] + " " # Append without hyphen
40                     else:
41                         word_cluster = word_cluster.strip() + "-" + ones[group] + " " # Append with hyphen for numbers 21-99
42                 else:
43                     word_cluster += ones[group] + " " # For numbers between 10 and 19
44
45             word_cluster += thousands[thousands_index] + " "
46             word = word_cluster + word
47
48             thousands_index += 1
49
50     return word.strip()
51
52 def main():
53     num = input("Number: ")
54     output = numtoword(num)
55     print(f'Converted Word: "{output}"')
56
57 main()
```

You may also see the full code on GitHub:

<https://github.com/Pr1ncei/Proyekto/blob/main/Projects/Python/numbertoword.py>